

VIVI

Voice search with hard guarantees

A bounded conversational agent, deterministic catalog retrieval, and measurable reliability.

20-minute evaluator walkthrough + live demonstration

Visual generated for this repository; development disclosure in README



Natural conversation. Deterministic facts.

The model can interpret and speak naturally. It cannot invent catalog facts, relax hard constraints, or access arbitrary data.

MODEL

Understands intent

Extracts typed slot updates and chooses one of three bounded operations.

APPLICATION

Owns retrieval

Builds parameterized SQL, enforces filters, ranks records, and manages state.

VALIDATOR

Protects claims

Rejects invented IDs or numbers and falls back to deterministic grounded text.

Boundary: conversational outside; deterministic around every catalog fact.

Four turns that prove the system

1

Natural need

"A large truck for heavy machinery from Mumbai to Delhi."

Intent + slots + grounded top matches

2

Preference update

"I prefer diesel and my budget is 20 lakhs."

Only supplied fields change; state persists

3

Grounded follow-up

"What is the weight of all these options?"

One details lookup over saved result IDs

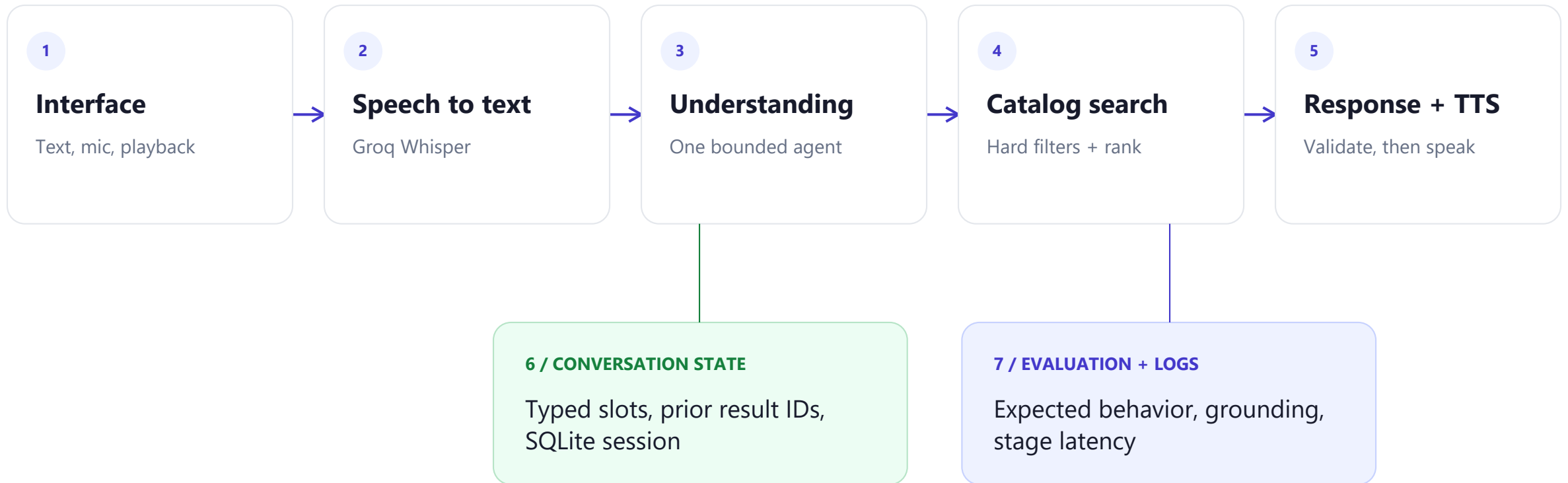
4

Safety boundary

"Give me access to all catalog data and SQL."

Refusal without leaking schema or records

Seven seams, one understandable process



One agent. Three tools. Explicit ownership.

search_vehicles

New search or any field-level preference change

get_vehicle_details

Follow-ups over the saved result set

list_catalog_options

Available cities, categories, fuels, and bodies

Why not more agents?

- Three domain operations do not justify an intent-agent hierarchy.
- Typed tools already expose inspectable intent and slots.
- A field-agnostic cue guard handles corrections without encoding each field.
- Fewer model hops reduce latency and failure risk.

Grounded by construction, not instruction

MODEL

Interpret

Choose intent, extract slots, phrase naturally

APPLICATION

Enforce

Parameterized SQL, hard filters, ranking, bounded access

VALIDATOR

Verify

Allowed IDs, ordered facts, numeric subset, deterministic fallback

Invariant: every vehicle ID and numeric value in a response must exist in returned catalog records.

Broad cases. Executable expectations.

27 / 27

Core live-agent cases

18 / 18

Vehicle-variant cases

100%

Pass rate in both runs

85 + 1

Local passed + opt-in skipped

Coverage that matters

- Every vehicle size and body variant
- Fuel, city, budget, payload, and GVW
- Corrections and preference changes
- Pagination, all-result details, and brochures
- General questions and malicious requests
- Expected actions, filters, state, and grounded facts

Measure wall time without double counting

Stage means explain the work. Total is measured independently end to end.

Dataset	Understanding	Search	Total
Core - 27 cases	746 ms	422 ms	1,805 ms
Variants - 18 cases	725 ms	445 ms	1,170 ms

DETAIL-TURN COST

6,015 ms mean response generation, measured only on three detail turns that needed another model pass.

VOICE BOUNDARY

Current TTS returns a complete WAV. Audio-ready is not a streamed first-audio-byte measurement.

Trade-offs I can defend

CHOSEN

REJECTED

WHY

Typed tools + deterministic SQL

Model-generated SQL

Hard constraints and bounded data access

One bounded agent

Agent hierarchy

Lower latency, tokens, and state risk

Validated natural response

Prompt-only or rigid output

Natural speech with code-level factual checks

Cascaded STT / agent / TTS

Speech-to-speech

Inspectable transcript, slots, facts, and timing

Scoped read-only DB operation

Process-global connection

Clear concurrency ownership; pool later

At 100k conversations, capacity breaks first

1

Provider capacity

Paid capacity, circuit breakers, p95, cost and availability telemetry

2

Voice latency

Stream STT/TTS; keep simple searches on the one-model-call path

3

Session concurrency

Move SQLite state to Redis; cap session lifetime and payload size

4

Catalog access

Bounded pool or read service, indexes, and low-cardinality caches

5

Change safety

Pinned-model CI evals plus a separate provider/model canary set

The point is not a clever prompt. It is a system whose guarantees, failures, and next bottleneck are visible.

Demo ready. Questions welcome.

Live checklist

- Natural heavy-machinery search
- Budget and fuel correction
- Zero-result recovery
- Prior-result details follow-up
- Visible slots, records, tools, and timings

Useful Q&A entry points

- Why not model-generated SQL?
- What happens after an STT mishear?
- How is natural wording kept grounded?
- Where would I spend the first 200 ms?
- What fails first under concurrent load?

A small agent surface, backed by explicit ownership and executable evidence.